GRASP: A Greenwashing Recognition and Scoring Pipeline for Structured BRSR Disclosures in India

Yugchhaya Galphat, Purab Keshwani, Khushi Dayaramani,
Ronit Chugwani, and Karan Sarani

Computer Department, Vivekanand Education Society's Institute of Technology
Mumbai, India

yugchhaya.dhote@ves.ac.in; 2023.purab.keshwani@ves.ac.in
2023.khushi.dayaramani@ves.ac.in; 2023.ronit.chugwani@ves.ac.in
2023.karan.sarani@ves.ac.in

Abstract

The Business Responsibility and Sustainability Report (BRSR) brings sustainability information into a common format for India's listed companies. Reviewing these reports requires analysts to connect written claims with filed figures, reporting periods, and industry context. GRASP organises this task through a seven-part pipeline. Structured digital filings (XBRL) are collected and stored with company and period details. Filing tags and concept mappings provide training labels. A BERT model adapted to BRSR text classifies passages under nine principles. Word lists and sentence similarity identify vague statements, while numerical and assurance checks produce linked review records. Principle mapping summarises the model's support for each reporting principle. A four-part Transparency Score supports company and sector comparisons, and an application programming interface provides access to the outputs. The saved results cover 1,164 companies. On 6,201 test passages from companies excluded from supervised training, the classifier agrees with filing-derived labels in 93.00% of cases. The Transparency Score correlates with 20 historical human ratings at $r = 0.6314$. Linked passages, score breakdowns, and peer comparisons help analysts follow the evidence and prioritise disclosures for review.

Keywords: BRSR; sustainability disclosure; weak supervision; BERT; vagueness detection; transparency scoring; sector benchmarking.

1 Introduction

India's Business Responsibility and Sustainability Reporting framework gives environmental, social, and governance disclosures a common structure. A 2021 circular from the Securities and Exchange Board of India (SEBI) made BRSR reporting mandatory for the top 1,000 listed companies by market capitalisation from financial year 2022–23. The framework covers nine principles of responsible business conduct. Essential indicators are mandatory, while leadership indicators are voluntary [1]. A common format helps organise reports, but cannot establish whether every response is complete or correct.

A data analyst must complete several manual steps when reviewing a filing. First, a passage must be found and assigned to a topic. Next, a relevant numerical value must be located and checked against the reporting year, unit, and group covered. Comparing companies adds another challenge

because business activities and applicable reporting items differ. Repeating these steps across hundreds of filings takes substantial time. For example, a statement that *water use decreased* needs a matching water measure, business boundary, and comparison period before the claim can be assessed.

Automated text analysis can reduce this workload. ClimateBERT identifies climate-related language and selected disclosure categories. ChatReport retrieves report passages and answers reporting questions with source links. ESGReveal extracts structured information from narrative reports [2, 3, 4]. These methods help organise large report collections. Three questions remain when building a connected workflow for Indian structured filings.

The first question is *how to connect a claim with evidence*. A phrase detector can find a promise, and a text-similarity model can identify an environmental topic. A supporting figure must also match the claim's meaning, period, and reporting context. The second question is *how to organise training labels*. Filing tags and concept maps provide a repeatable assignment procedure, with the label source recorded for each passage. The third question is *how to interpret a company score*. The calculation, included items, sector group, and reference source provide this context. GRASP brings these requirements together in a connected BRSR workflow with clearly defined tests [5, 6].

GRASP addresses the first question by storing narrative passages and structured facts under the same filing. Language checks flag passages for review, while numerical and assurance checks keep the linked sentence and reported item available for inspection. The second question is addressed by recording three label sources and comparing results by source, company, and excluded question concept. The third question is addressed by showing four score components and comparisons within recorded sectors. The contribution combines links to filed evidence, filing-derived training labels, detailed evaluation, and a score with a visible breakdown.

The study provides a 42,409-sample dataset with documented weak labels and separate companies for supervised training, validation, and testing. The classifier is assessed by label source and on concepts excluded from training. The language detector is evaluated with fixed settings, while the numerical checks supply linked review records. Company-score results move from one worked example to the full financial-sector group and all recorded sectors. The evaluation identifies the data group and reference source for each model result, human-score comparison, and service check.

The remaining sections of this paper follow the workflow from earlier research to practical use. Section 2 reviews related methods and Indian disclosure studies. Section 3 presents the proposed solution and the flow from filing data to review outputs. Section 4 explains the seven modules and score calculations. Section 5 defines the test data and reports the audited model results. Section 6 presents the four reporting levels, from one company to all sector groups. Section 7 discusses output interpretation and practical use. Section 8 summarises the findings and future work.

2 Literature Review

Automated sustainability analysis includes several tasks: finding relevant passages, identifying commitments, checking whether a statement is specific, retrieving evidence, and organising results. These tasks provide useful signals, but a language signal alone does not prove that a company has misled a reader. Calamai et al. review these tasks and identify continuing problems with available datasets and evaluation methods [5]. The review below keeps each task separate from a claim of verified greenwashing.

2.1 Language models and training labels

BERT learns word meaning from the surrounding text and can then be trained for a labelled classification task [7]. Gururangan et al. show that further training on domain and task text can

improve several classification tasks [8]. This finding supports adapting a model to sustainability reports. Lower language-model loss, however, does not alone establish better principle classification. Sentence-BERT represents each sentence as a numerical vector, allowing sentence meanings to be compared with cosine similarity [9]. Similarity helps find related language, but does not confirm that a statement is supported by evidence.

Weak supervision uses rules to reduce manual labelling. Snorkel combines labelling rules and estimates uncertainty in the resulting labels [10]. GRASP uses a simpler fixed assignment from filing tags and concept names. An incorrect assignment can therefore repeat across many examples. Kapoor and Narayanan explain how data leakage and weak evaluation design can make machine-learning results appear stronger than the evidence supports [11]. This motivates checks on company overlap, repeated text, label sources, and saved-model loading.

2.2 Climate topics, commitments, and communication quality

ClimaText provides labelled sentences for climate-topic detection [12]. A topic label helps select relevant text, but does not assess the support for a claim. ClimateBERT adds pretraining on climate text and evaluates climate-language tasks [2]. Bingler et al. use ClimateBERT to study selective and unclear reporting under the Task Force on Climate-related Financial Disclosures (TCFD) framework [13]. A later study relates climate-related cheap talk, meaning statements with limited substance, to climate initiatives, emissions growth, and reputation risk [14]. These studies already connect corporate language with performance-related information. The contribution of GRASP should therefore be stated for the specific BRSR workflow.

ClimateBERT-NetZero uses approximately 3,500 expert-labelled examples, target classification, and question answering to study net-zero and reduction commitments [15]. Finding a target and extracting the target details are separate tasks. Vinella et al. use generated labels and a fine-tuned ClimateBERT model to study greenwashing risk [16]. Generated labels increase the available training data, but the evaluation still depends on how those labels were produced. The same concern applies to AI-assisted concept mapping in GRASP.

2.3 Report-level tools and Indian disclosure studies

ChatReport combines language-model assistance, report passages, and expert feedback to support disclosure review [3]. ESGReveal uses document preparation, standard questions, and language-model agents to extract structured information from ESG reports [4]. Both studies support the value of a connected review tool. GRASP starts from structured XBRL facts, where values already have filing identifiers and numerical context. For these available fields, page-layout extraction is not required.

Shankar and Xu describe a BERT sentence classifier trained with expert labels for Indian sustainability reports [17]. The source is an SSRN working paper posted in 2025, with a manuscript dated 2024. Garg et al. review BRSR disclosures from 1,012 Indian businesses for 2022–23 [6]. This institutional report provides Indian reporting context and a broad review of disclosure practices. The report is not a machine-learning detection benchmark.

2.4 Design choices for the BRSR workflow

The literature supports three choices: connect evidence to filing identifiers, record how training labels are obtained, and show how a company score is calculated. GRASP combines these choices through a filing database, a nine-principle classifier, review flags, and sector comparisons. The evaluation compares label sources, principle categories, and company groups. Table 1 summarises

selected modelling and disclosure studies in year order and explains the design lesson from each source.

Table 1. Related research and lessons for GRASP. Each study uses a different task and dataset, so reported findings are not scores on one shared test.

Year	Objective	Methodology	Main findings	Scope / limitation	Inference drawn
2019	Learn general text features: BERT [7].	Pretraining with context on both sides of a word, followed by labelled training.	One pretrained model supports several language tasks.	General pretraining does not define BRSR categories.	Add a nine-principle output layer and test the BRSR task.
2019	Compare sentence meanings [9].	Paired sentence encoders and vector similarity.	Supports efficient search for related sentences.	Similar wording does not prove factual support.	Use similarity as a review signal.
2020	Adapt a model to specialised text [8].	Further pretraining on domain and task text.	Improves several reported classification tasks.	Gains depend on the training text and target task.	Record the adaptation text separately from the labelled split.
2020	Find climate-topic sentences: ClimaText [12].	Dataset with sentence-level climate labels.	Provides data for testing topic detection.	A topic label does not assess a claim's credibility.	Separate topic detection from evidence checks.
2021	Model climate language: ClimateBERT [2].	Further pretraining on climate paragraphs.	Improves the reported climate-language tasks.	Climate tasks do not cover every social or governance principle.	Test adaptation on the intended task.
2022	Study climate-risk reporting [13].	ClimateBERT analysis of TCFD-related language.	Finds vague claims and selective reporting.	TCFD categories differ from BRSR principles.	Keep labels specific to the reporting framework.
2023	Find net-zero and reduction targets [15].	Expert labels, ClimateBERT, and question answering.	Finds targets and related details.	Detecting a target does not verify delivery.	Separate commitments from measured outcomes.
2023	Support report review: ChatReport [3].	Passage retrieval, model responses, and expert feedback.	Links report analysis to source passages.	Retrieved passages and answers still need evaluation.	Show the source text with each output.
2024	Study risk using generated labels [16].	Generated labels and ClimateBERT fine-tuning.	Demonstrates an early language-model approach.	Results depend on the generated-label rules.	Identify AI labels separately from human labels.
2024	Compare cheap talk with company outcomes [14].	Text-based indicator and outcome analysis.	Finds links with emissions growth and negative coverage.	A statistical link does not verify every claim.	Check the context of numerical evidence.
2025	Extract ESG report data: ESGReveal [4].	Document preparation, standard queries, retrieval, and model agents.	Supports structured extraction and disclosure analysis.	Results depend on documents and model choice.	Use available structured facts before page extraction.

Continued on next page

Table 1 continued

Year	Objective	Methodology	Main findings	Scope / limitation	Inference drawn
2025	Classify Indian report sentences [17].	Expert-labelled text and supervised BERT.	Reports sentence-level greenwashing detection.	Working paper with different data and labels.	Avoid direct accuracy comparisons across studies.
2025	Review Indian BRSR disclosures [6].	Institutional review of 1,012 businesses for 2022–23.	Describes BRSR reporting practices.	A disclosure study, not a detection test set.	Consider Indian reporting conditions in comparisons.
2025	Review greenwashing text research [5].	Survey of tasks, data, and methods.	Identifies separate tasks and testing gaps.	No single language task establishes greenwashing.	State exactly what each output measures.

3 Proposed Solution

GRASP organises sustainability review around a shared source: the company's structured BRSR filing. Narrative passages and numerical facts are stored together with company and reporting details. An analyst can therefore move from a flagged statement or a score component to the underlying disclosure. The central idea is to connect language analysis with reported evidence and present the resulting records in one review workflow.

The design responds to the three questions introduced in Section 1. Evidence links connect a passage with reported values. Recorded label sources explain how training examples receive principle assignments. A visible score breakdown and sector comparison explain a company's position. The contribution brings these functions together for Indian BRSR filings, with a defined role for each model, rule, and calculation.

The seven modules shown in Figure 1 follow this design. Filing acquisition and storage create the common data record. Narrative selection removes unusable text and derives weak training labels from principle tags and concept mappings, including the documented AI-assisted assignments. Domain adaptation prepares BERT for BRSR language, and supervised training produces a classifier for the nine reporting principles.

The analysis modules use the stored passages and facts for different purposes. Word lists and sentence similarity identify vague language. Numerical and assurance checks create separate review records containing the relevant claim and reported item. Principle mapping uses the classifier's probabilities to summarise support for each principle; a zero-shot method provides the evaluation comparator. The Transparency Score combines Completeness, Specificity, Clarity, and mapped SDG breadth, followed by comparisons within recorded sectors. Principle probabilities and numerical review flags remain separate from the four score components.

The service layer makes company records, review flags, principle summaries, score breakdowns, and peer comparisons accessible through an API. Results can be read at four levels: one company, selected companies within a sector, the complete financial-sector group, and all recorded sectors. Available paired filings also support year-by-year comparison. Section 4 explains the internal working of each module, while Section 5 defines the evidence used to assess the outputs.

4 Methodology

GRASP connects seven modules through a shared filing record. Figure 1 uses the same titles as the following subsections. Filing collection supplies the data, model training produces the classifier, and three analysis branches provide service outputs. The arrows show the main routes; the module descriptions explain other database inputs and shared calculations. The Transparency Score uses reported-item coverage, evidence markers, vagueness flags, and a fixed SDG mapping. Principle probabilities and claim–figure flags remain separate outputs.

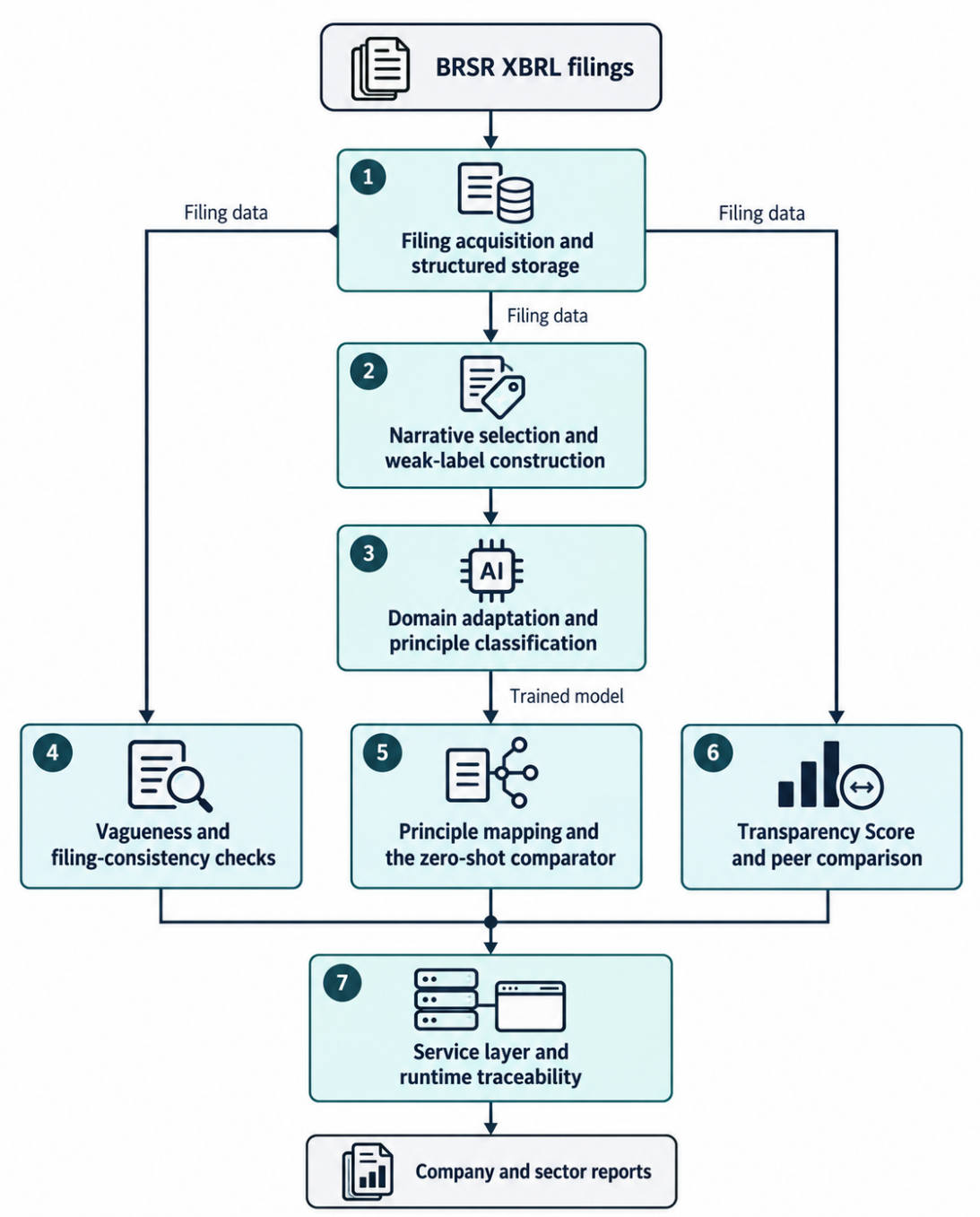


Figure 1. GRASP workflow from BRSR filings to company and sector reports. Numbered modules correspond to the methodology subsections; arrows show the main data and model routes.

4.1 Filing acquisition and structured storage

The input consists of BRSR filings in eXtensible Business Reporting Language (XBRL), together with company identifiers. A download log records completed and failed files so collection can resume after an interruption. The parser reads structured filing data directly. A concept names a reporting item, such as water withdrawal. The context records the company, period, and any reporting categories, while the unit records how a numerical value is measured [18]. This structure keeps a figure linked to the information needed for comparison.

The database has three linked record types. A company record stores the Corporate Identity Number (CIN), name, and industry code. A filing record stores the company link, reporting dates, and source location. A fact record stores the concept, original value, unit, stated precision when available, period, and dimensions. Dimensions are categories such as gender or employment type. A workforce count can therefore retain these categories instead of becoming one combined number. Values are stored as text and converted to numbers only when a calculation requires arithmetic. A company–year uniqueness rule limits duplicate ingestion of the same filing.

The output is a searchable database of reported facts. Each value retains a link to the source filing, reported concept, period, and unit. These links support later passage review, numerical comparison, and company-score calculation.

4.2 Narrative selection and weak-label construction

Text candidates come from concepts containing `ExplanatoryTextBlock`. Before counting words, the dataset builder removes URLs. At least 15 words of useful content must remain. Bare links, short fragments, and listed non-answers are removed. These rules define the classifier dataset. The company-scoring modules use different text-length rules, explained in Section 4.6. Classifier passages and scoring passages are therefore different groups of text.

Principle labels come from three sources, checked in order. An explicit `PrincipleAxis` tag supplies a Section B label. If that tag is unavailable, a Section C concept-to-principle map supplies the label. A numbered `NotesPrincipleX` concept provides the third source. The extension includes 29 AI-assisted assignments, with existing configured mappings taking priority when maps overlap. The process produces taxonomy-derived weak labels rather than independent passage-level expert judgements; the AI-assisted assignments have no completed independent expert verification.

Candidates are grouped by company, concept, and cleaned text. Repeated entries with the same label produce one row. A group with several labels is removed from the single-label dataset. These groups can include passages covering several principles or generic text reused under different questions. The resulting dataset contains 42,409 samples. The CSV records the label source for analysis, while only the passage text enters the classifier.

The labelled dataset is split by CIN. Training contains 29,834 passages from 814 companies, validation contains 6,374 passages from 175 companies, and test contains 6,201 passages from 175 companies. No company appears in more than one split. This company-based partition defines the main supervised evaluation. Exact wording shared across different companies is counted separately in the text-overlap check.

4.3 Domain adaptation and principle classification

The encoder starts from `bert-base-uncased`. Masked language modelling adapts BERT to sustainability text: selected word pieces are hidden, and the model learns to predict the missing text. A nine-output classification layer is then added, and labelled training adjusts the model for the nine principles. For a passage x, the model returns a probability $p_k(x)$ for each principle k.

The largest probability gives the Top-1 prediction. The two largest probabilities give the Top-2 candidate list [7, 8].

Model development and component roles. The final model follows four steps: base BERT, adaptation to BRSR text, supervised training for nine principles, and recovery of the selected seed-99 checkpoint. A checkpoint is a saved set of model parameters. Adaptation changes the parameters while keeping the BERT structure. Fine-tuning adds the classification output layer and updates the encoder. Recovery corrects saved parameter names and verifies loading without further training. MPNet supplies sentence similarity for vagueness checks and the zero-shot comparison. The final audited evaluation uses the recovered BERT classifier; earlier ClimateBERT and ensemble runs belong to the passage-level experiment history.

The classifier accepts at most 512 tokens, meaning word pieces produced by the tokenizer. Dynamic padding adds only the padding needed within a batch. Training processes four passages at a time and combines gradients across four such microbatches before updating the model. An epoch is one pass through the training set. Fine-tuning runs for five epochs with learning rate 2×10^{-5}, weight decay 0.01, 300 warmup steps, and cosine learning-rate decay. The learning rate controls update size, warmup raises the rate gradually, and cosine decay then lowers the rate along a smooth curve. Weight decay limits parameter growth.

Cross-entropy loss penalises low probability for the reference label. Class weights use the square root of inverse training frequency. Less common classes receive more weight, but the square root limits the increase. A random seed makes the training choices repeatable. Five seeds, 42, 1, 7, 99, and 2024, were compared using validation macro-F1. Seed 99 and checkpoint 7,460 form the audited final classifier.

Checkpoint recovery verifies the parameters used for inference. The selected file contains 50 LayerNorm parameters with older `gamma`/`beta` names. Recovery changes these names to `weight`/`bias`, requires every parameter to load, and confirms tensor equality after saving and reloading. The test result uses this recovered model. The earlier masked-language-model builder reads text across the company database, and checkpoint-specific adaptation membership has not been verified. The recorded company-separation assertions therefore apply to labelled fine-tuning.

4.4 Vagueness and filing-consistency checks

The risk module produces two kinds of review output. The language branch combines a hedging word list with sentence-similarity checks. The numerical branch produces claim–figure, assurance-gap, and omission records with links to the relevant filing material. Section 5 reports the evaluation group for each measured component.

4.4.1 Hedging words and evidence markers

A hedging word list identifies promises, conditional commitments, unclear time expressions, and progress claims without a measure. Evidence patterns look for percentages, currency amounts, physical quantities, workforce or incident counts, reporting years, named standards, and selected formal governance systems. The code names this marker `has_kpi`. This marker includes numerical indicators, named policies, and standards. Specificity records the presence of these details in a passage.

The word-list rule flags a passage containing hedging without an evidence marker. For passages longer than 80 words, hedging density, calculated as matches divided by word count, must also reach 0.005. A separate rule flags bare URLs and listed website redirects. The final pattern set requires digits in currency matches, includes MW/GW units and statutory terms, and handles URL redirects explicitly.

4.4.2 Comparing sentence meanings

The similarity check uses the `all-mpnet-base-v2` sentence encoder [9, 19]. Each passage is split into sentences. Sentence vectors are compared with two fixed reference sets: vague statements and specific statements. Cosine similarity measures how close two vectors are. Let v_j be the largest similarity between sentence j and a vague reference, and q_j the largest similarity to a specific reference. The rule selects the sentence with the largest $q_j - v_j$, treating that sentence as the most specific part of the passage.

The decision uses the selected sentence. Without an evidence marker, the similarity rule requires $v_j \geq 0.42$ and $v_j > q_j$; either the word-list rule or this similarity rule can flag the passage. With an evidence marker, flagging requires $v_j \geq 0.58$ and $v_j - q_j \geq 0.15$. A URL-redirect match takes priority. The decision therefore uses the most specific sentence together with the evidence-marker status.

The freeze file records the detector version and thresholds before the audited run. The settings remain fixed during evaluation. Batches 1 and 2 form the 80-example development group.

4.4.3 Claim–figure, assurance, and omission checks

The numerical branch checks selected topics, including emissions, water, waste, and non-renewable energy. A sentence must contain both improvement wording and a phrase linked to the measure. A review record is created when the linked filed value increases despite reduction-style wording. The record keeps the sentence, concept, previous and current values, and assurance information. Matching within a sentence reduces accidental links, such as comparing a Scope 2 reduction claim with a Scope 1 increase.

The assurance-gap rule links a domain-specific fact with reported assurance information. The omission rule identifies a prior-period concept without a matching current-period concept. Analyst review compares measure definitions, total versus per-output quantities, units, periods, and business boundaries using the linked filing context.

The current assurance-gap implementation selects a domain-matched fact without fully restricting period, unit, and dimension, and treats a text value of zero as absent. The omission rule excludes selected zero-valued prior entries. These selection rules can affect review flags, so context review inspects the original values and reporting categories. The Completeness calculation separately accepts structured zeros as valid responses. Numerical records are reported as rule-generated review signals; the accuracy results in Section 5 cover the language branch.

4.5 Principle mapping and the zero-shot comparator

The mapping service reuses the recovered classifier. The output layer produces nine raw scores, called logits. Softmax converts these scores into principle probabilities. Reuse of the trained classifier is called supervised transfer in the evaluation. For passages filed under principle k, the service averages $p_k(x)$ and keeps the lowest-scoring passage for inspection. The company summary averages the available principle means. This value describes model support for the recorded principle assignment.

The zero-shot comparison uses MPNet vectors for the same test passages. Zero-shot means that this ranking method is not fitted to the principle-labelled training set. Each principle has an expanded definition and a fixed set of example passages, called anchors. The ranking score combines similarity to the definition and to the anchors:

$$z_k(x) = 0.50\, d_k(x) + 0.50 \left[0.70 \max_a s_{ka}(x) + 0.30\ \text{mean}_a\, s_{ka}(x)\right], \tag{1}$$

where d_k compares the passage with the principle definition, and s_{ka} compares the passage with anchor a. Both methods rank nine principles on the same examples. The supervised classifier has access to training labels; the zero-shot ranking method does not. The observed difference therefore combines the effects of the model representation and labelled training.

4.6 Transparency Score and peer comparison

The Transparency Score combines four percentages on a 0–100 scale:

$$TS_f = 0.40\, C_f + 0.30\, Sp_f + 0.20\, Cl_f + 0.10\, G_f. \tag{2}$$

The implementation fixes the component weights at 0.40/0.30/0.20/0.10. Each component is rounded to two decimal places, and the final sum is also rounded to two decimal places. The resulting score summarises disclosure coverage, detected specificity, language flags, and mapped SDG breadth.

4.6.1 Completeness: which reporting items have a usable response?

Let U contain every distinct concept name in the saved database. Let A_f contain the concepts with at least one accepted response in filing f. Completeness is the share of that full concept list with an accepted response:

$$C_f = 100\, \frac{|A_f|}{|U|}. \tag{3}$$

The saved database contains 1,057 distinct concepts. This total, the denominator, includes both structured and narrative reporting items observed in the database. Repeated facts under several dimensions count each concept only once.

Blank responses and listed placeholders such as *NA* and *not applicable* do not count. Structured zeros do count. Narrative text blocks reject placeholders and a standalone zero. Text longer than 30 characters needs at least ten words separated by whitespace; shorter values can pass if no placeholder matches. Address fields use a separate rule. All companies share the full-database concept denominator, including sector-specific items. Comparisons under one score version keep this concept list fixed.

4.6.2 Specificity: how often does a narrative contain an evidence marker?

Scoring uses text-block facts with at least 30 characters. Duplicate concept–text pairs are removed after trimming extra spaces. The same wording under different concepts can still count as several passages. Let N_f be the number of retained passages and K_f the number with at least one evidence pattern from Section 4.4. Specificity is

$$Sp_f = 100\, \frac{K_f}{N_f}. \tag{4}$$

A paragraph with five numbers counts once, as does a paragraph with one recognised standard. The denominator is the number of passages. The resulting percentage gives the share of selected text containing at least one marker from the configured pattern set.

4.6.3 Clarity: how much selected text avoids a vagueness flag?

Let V_f be the number of scoring passages flagged by the combined fixed language detector. Clarity is

$$Cl_f = 100 \left(1 - \frac{V_f}{N_f}\right). \tag{5}$$

Clarity gives the percentage of selected passages without a vagueness flag. When no scoring passages exist, the code returns zero for both Specificity and Clarity.

4.6.4 SDG breadth: how many goals are linked to marked passages?

The fourth component refers to the 17 United Nations Sustainable Development Goals (SDGs) [20]. A principle qualifies when an explicitly tagged narrative has at least 20 characters and an evidence marker. A fixed map then links that principle to goal numbers. The map uses P1: {16}; P2: {9,12}; P3: {3,5,8}; P4: {10,17}; P5: {8,10,16}; P6: {6,7,12,13,15}; P7: {16,17}; P8: {1,8,10,11}; and P9: {9,12}. For qualifying principles D_f and mapping M, the score counts distinct linked goals:

$$G_f = 100\,\frac{\left|\bigcup_{k\in D_f} M(k)\right|}{17}. \tag{6}$$

Saved exports name this component *SDG Alignment*; the calculation measures mapped goal breadth. The map covers 14 of the 17 goals and has a maximum value of 82.35%. The goal-level calculation uses the fixed principle map rather than checks against the 169 individual targets. Duplicate removal keeps one principle for each concept–text pair according to database record order.

4.6.5 Sector grouping and percentiles

The benchmark links each company score to a sector using the recorded industry code and a local mapping. The financial analysis uses the saved group *Financial Services (excl. Insurance)*. Insurance and auxiliary financial services remain separate groups. These recorded labels define the peer sets used throughout the results.

For company c in sector S, the percentile is

$$P_c = 100\,\frac{\#\{j \in S : TS_j < TS_c\}}{|S|}. \tag{7}$$

The percentile gives the percentage of sector companies with a strictly lower score. Tied scores receive the same percentile, and the highest value need not equal 100. The main sector comparison includes groups with at least five companies. Sector means, medians, ranges, and standard deviations describe the scores within these recorded groups.

4.7 Service layer and runtime traceability

The service provides company search, filing analysis, score breakdowns, peer comparisons, and principle mapping. The principle-mapping endpoint returns the method name and model path so an operator can identify the classifier used. The verified route reports `recovered_seed99_supervised_classifier` and uses the recovered checkpoint from Section 4.3. If that classifier is unavailable, the endpoint returns an error. A different method is therefore not silently presented under the same label.

The successful request in the audit records model loading for one company. The returned mean probability summarises that filing. Model-loading status, passage probabilities, and peer comparisons have separate response fields.

5 Evaluation Design and Audited Model Results

The evaluation reports four kinds of output: agreement with filing-derived labels, performance against development labels, comparison with historical human ratings, and saved company scores.

Final values come from the consolidated audit and recovered-result JSON files. Each comparison records the dataset, reference source, and unit of analysis.

5.1 Data groups, evaluation measures, and uncertainty

For principle classification, Top-1 agreement is the share of passages whose highest-ranked principle matches the recorded label. Top-2 agreement is the share whose recorded label appears among the two highest-ranked principles. Top-2 gives an analyst a short candidate list. A higher Top-2 value does not mean better accuracy when only one answer is allowed.

Precision shows how reliable a predicted category is. If 100 passages receive an environmental-principle prediction and 90 carry that reference label, precision is 90%. Recall shows how much of a reference category is found. If 100 passages carry the environmental label and the classifier finds 80, recall is 80%. F1 balances precision and recall through the harmonic mean:

$$F1_k = \frac{2P_kR_k}{P_k + R_k}, \qquad \text{MacroF1} = \frac{1}{9}\sum_{k=1}^{9} F1_k. \tag{8}$$

Macro-F1 gives each principle equal weight, so the largest class cannot determine the entire score. Support means the number of reference examples for a principle. The references are weak labels. A result described as correct therefore means agreement with the recorded assignment, not an independent judgement of the passage's meaning.

A 95% confidence interval shows uncertainty in an estimate under the stated sampling method. For classification and mapping, the audit uses 2,000 bootstrap samples with seed 4206. Companies are sampled with replacement, and all passages from a selected CIN are kept together. This CIN-cluster method accounts for passages from the same company being related. Paired comparisons use the same sampled companies for both methods. These intervals describe variation within the available data; label errors and earlier exposure during model development remain outside that estimate. The 20-company score comparison uses 10,000 bootstrap samples of company-score pairs with seed 42. No detector intervals were calculated.

5.2 Method comparisons and generalisation checks

Table 2 summarises three comparisons: results by label source, performance on concepts excluded from training, and supervised versus zero-shot mapping. The table records the examples and training conditions for each comparison. These are method and data-split comparisons rather than a controlled change to one component.

The reported comparisons use the final labelled split and the training conditions listed in the table. Domain adaptation is part of the final model's training sequence. No separate numerical gain is assigned to adaptation or data cleaning.

5.3 Dataset size and results by label source

The final single-label dataset contains 42,409 samples: 2,479 from explicit principle-axis facts, 36,513 from Section C mappings, and 3,417 from principle notes. The builder excludes 36,491 noisy facts and removes 2,640 groups with several labels. Fact rows and groups are different counting units. These exclusions cannot simply be added as final examples.

The Section B build starts with 41,898 narrative candidates. The content filter removes 16,684 candidates, leaving 25,214 clean entries. Removing 18 repeated same-label entries and 22,717 entries from groups with several labels produces 2,479 final rows. These steps reconcile the current source counts.

Table 2. Three comparisons of Top-1 agreement with filing-derived labels. Data and training conditions are recorded for each row. Percentages from different rows describe different groups. Sections 5.3, 5.5, and 5.7 give intervals and further measures.

Comparison	Evaluation population	Top-1 agreement	Interpretation
Label source	B: 352; C: 5,364; notes: 485 segments	B: 63.35%; C: 95.15%; notes: 90.72%	Results for parts of the same 6,201-row test set. Sources use different text and labelling rules.
Full-concept and excluded-concept training	Same 1,190 segments from 174 companies	Full concept: 97.98%; excluded concept: 87.31%	The training sets differ in concept coverage, example count, and seed.
Zero-shot MPNet versus supervised BERT	Same 6,201 segments from 175 companies	MPNet: 67.63%; BERT: 93.00%	The methods have different access to labels. The BERT value reuses the classifier result.

Table 3. Final weak-label dataset by principle. The last column gives example counts in the company-separated test set.

Principle	Dataset rows	Test rows
P1: Ethics and governance	5,015	773
P2: Sustainable goods and services	5,566	808
P3: Employee and worker wellbeing	11,835	1,715
P4: Stakeholder interests	3,128	469
P5: Human rights	3,095	449
P6: Environmental responsibility	5,749	823
P7: Public policy advocacy	543	69
P8: Inclusive growth	2,475	368
P9: Consumer responsibility	5,003	727
Total	**42,409**	**6,201**

Table 4 reports agreement of 63.35% for explicit-axis passages and 95.15% for concept-mapped passages. Section C supplies 86.1% of the full labelled dataset and 5,364 of the 6,201 test rows. The overall value therefore reflects the source mix. Section B contains the examples retained by the single-label filtering procedure. Figure 2 presents the three sources together, and the excluded-concept tests provide a second view of performance across reporting questions.

5.4 Recovered classifier performance and training history

The recovered seed-99 model matches 5,767 of 6,201 reference labels. Top-1 agreement is 93.00% (95% CI: 92.04–93.89%). The reference label appears among the two highest predictions for 6,057

Table 4. Recovered classifier results by label source on the same test set. A company can supply several sources, so the company counts overlap. Top-1 intervals are calculated by resampling companies.

Source	Segments	CINs	Top-1	Top-2	Top-1 95% CI
Section B: explicit axis	352	71	63.35%	80.40%	56.55–69.67%
Section C: concept map	5,364	174	95.15%	99.01%	94.41–95.92%
Principle notes	485	103	90.72%	95.46%	86.55–94.09%

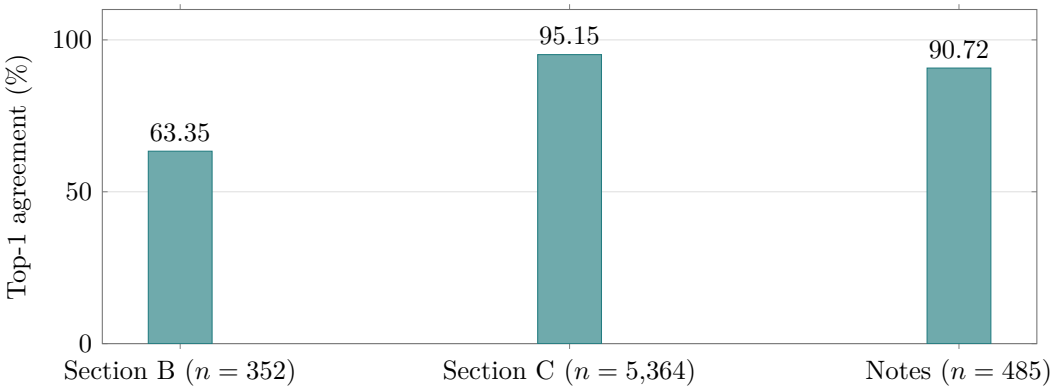
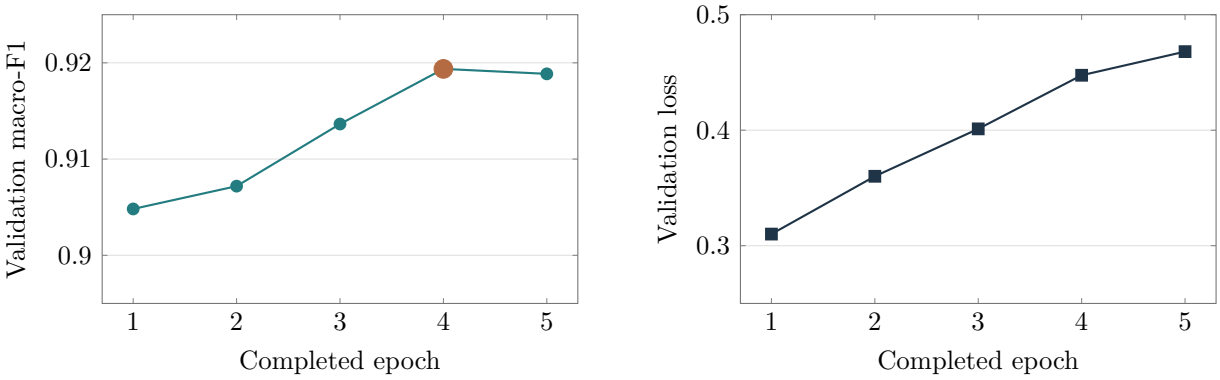


Figure 2. Principle-label agreement by source. The bars compare explicitly tagged passages, concept-mapped passages, and principle notes using the filing-derived reference labels. Table 4 gives confidence intervals calculated by resampling companies.



(a) Category agreement is highest at epoch 4.

(b) Loss records changes in prediction confidence.

Figure 3. Five recorded training epochs for seed 99. The marked point identifies checkpoint 7,460, selected by validation macro-F1. The two panels present category agreement and probability-based loss, showing the recorded basis for checkpoint selection.

examples, giving 97.68% Top-2 agreement (95% CI: 97.10–98.21%). Macro-F1 is 0.9158 (95% CI: 0.9036–0.9277). The text-overlap check identifies 50 exact train/test matches across different companies, or 0.81% of test rows. The supervised split has zero CIN overlap.

Figure 3 shows the five recorded epochs for seed 99. Validation macro-F1 rises from 0.9048 at epoch 1 to 0.9194 at epoch 4, then reaches 0.9189 at epoch 5. Validation loss rises over the same period. F1 describes category decisions, while loss also considers prediction confidence. The recorded selection criterion chooses epoch 4 by validation macro-F1.

Table 5 groups performance by principle; Table 4 groups performance by label source. Each principle row shows prediction reliability, reference-category coverage, and the balance between both. P7 has only 69 test examples, so the P7 estimate needs extra care. Figure 4 summarises F1, while the table also gives precision, recall, and example counts.

5.5 Tests on excluded concepts and multiple labels

Separating companies asks whether the model can classify passages from another company. Separating concepts asks whether the model can classify passages for reporting questions excluded from labelled training. The first check excludes 17 Section C concepts and evaluates 1,190 rows. On these same rows, the full-concept model reaches 97.98% Top-1 agreement and the excluded-concept model reaches 87.31%. The paired difference is −10.67 percentage points (95% CI: −12.36 to

Table 5. Results for nine principles on 6,201 passages. Precision shows how often a predicted principle matches the reference. Recall shows how much of the reference category is found. F1 balances both.

Principle	Support	Precision	Recall	F1
P1: Ethics and governance	773	0.9592	0.9431	0.9511
P2: Sustainable goods and services	808	0.9154	0.8837	0.8992
P3: Employee and worker wellbeing	1,715	0.9533	0.9644	0.9588
P4: Stakeholder interests	469	0.9611	0.9488	0.9549
P5: Human rights	449	0.9009	0.8508	0.8751
P6: Environmental responsibility	823	0.8943	0.9356	0.9145
P7: Public policy advocacy	69	0.8551	0.8551	0.8551
P8: Inclusive growth	368	0.8921	0.9212	0.9064
P9: Consumer responsibility	727	0.9259	0.9285	0.9272
Macro average	6,201	0.9175	0.9146	0.9158

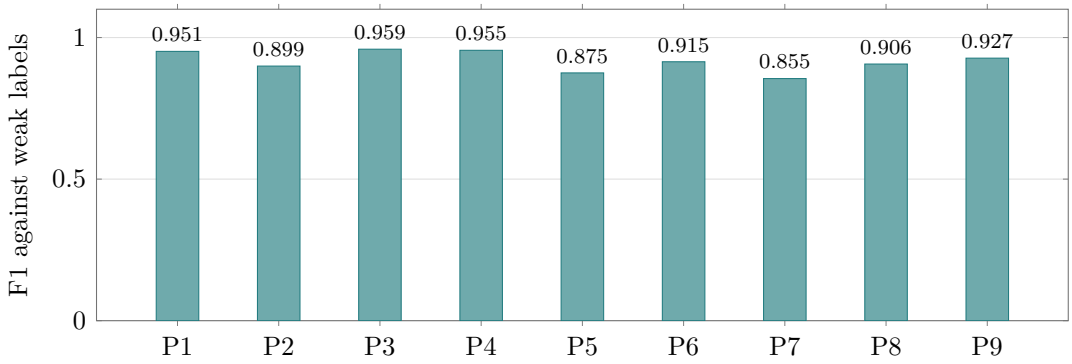


Figure 4. F1 by principle for the recovered classifier. A higher bar means a better balance of precision and recall against the recorded weak labels. P7 has 69 test examples, compared with 1,715 for P3. Table 5 gives all example counts and measures.

−8.96). Both checkpoints passed strict parameter-name recovery before re-evaluation.

The full-concept model uses seed 99. The excluded-concept model uses seed 42 and fewer training examples. The paired interval describes the difference on shared test rows under these training conditions. The comparison includes changes in concept coverage, training examples, and seed.

Across three excluded-concept runs, mean Top-1 agreement is 81.64%, with sample standard deviation 4.92 percentage points. Mean Top-2 is 95.34%, with standard deviation 1.31 percentage points. Mean macro-F1 is 0.6752, with standard deviation 0.0170. Figure 5 shows each run and test size. Standard deviations describe variation across the three runs. All runs reuse the original test-company group, and concept exclusion is defined for labelled training. The earlier adaptation-corpus membership is described in Section 4.3.

The multi-label version groups company text and combines the recorded principle labels for each group. The test set contains 5,934 examples from the same 175 companies. Hit@1 checks whether the highest-ranked prediction belongs to the recorded label set and reaches 93.36% (95% CI: 92.54–94.11%). At threshold 0.5, micro-F1 is 0.8898 (0.8743–0.9035), and macro-F1 is 0.8782 (0.8606–0.8942). The single-label model reaches Hit@1 of 93.55% on the same grouped labels. This comparison evaluates an alternative target definition: one company-text example can contain several principle assignments, including assignments from generic text reused under different questions.

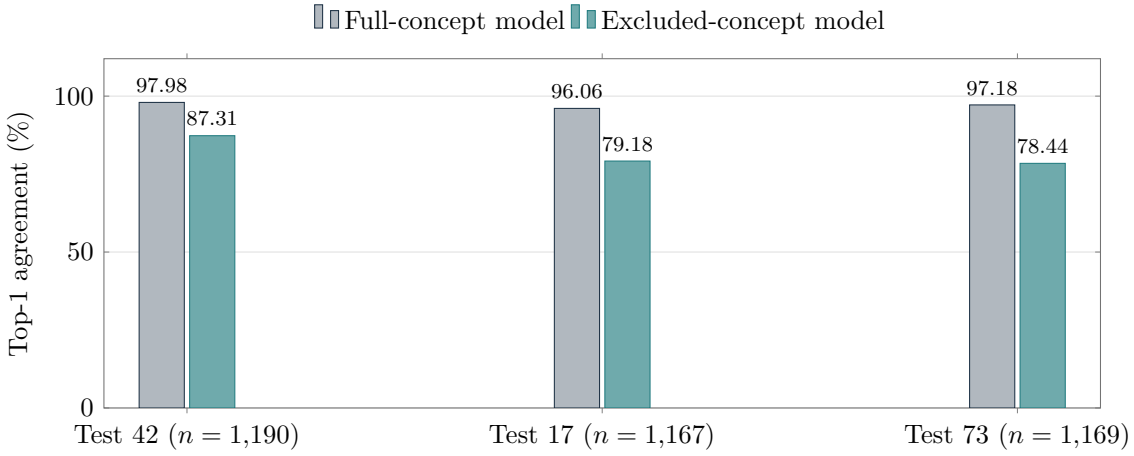


Figure 5. Full-concept and excluded-concept training on three shared test groups. The tests contain 174, 173, and 174 companies from the original split. Excluded-concept mean agreement is 81.64%, with sample standard deviation 4.92 percentage points across runs. Training examples and seeds are described in Section 5.5; the standard deviation summarises variation across these runs.

Table 6. Fixed text-vagueness detector on two label groups. The first row uses 80 development examples, and the second uses 70 AI-labelled examples. Reference conditions are stated in the evaluation text.

Reference labels	N	Accuracy	Precision	Recall	F1
Historical development labels	80	0.7500	0.7073	0.7838	0.7436
AI annotations	70	0.7286	0.6000	0.8077	0.6885

5.6 Frozen text-vagueness results

Table 6 reports the fixed detector on two separate groups. Batches 1 and 2 contain 80 earlier development examples. The detector flags 41 examples, and 29 flags match the reference vagueness label. The detector finds 29 of the 37 reference-vague examples. These counts give precision 0.7073 and recall 0.7838. In total, 60 of the 80 decisions match the reference, giving accuracy 0.7500 and F1 0.7436.

Batch 3 contains 70 AI-labelled examples. The detector flags 35 passages, and 21 flags match the AI vagueness label. The detector identifies 21 of the 26 passages labelled vague, giving F1 0.6885. Figure 6 displays precision, recall, and F1 alongside the 80-example development group. The groups use different passages and reference sources and are reported separately.

The reported precision, recall, and F1 cover the combined word-list and sentence-similarity detector. Claim–figure and assurance-gap outputs are rule-generated review records without a completed accuracy evaluation. Batches 1 and 2 are development data, and annotation blinding is not documented in the saved records. Batch 3 uses one AI-generated annotation set and is an exploratory reference comparison. Continuous-score ROC AUC and between-rater agreement are outside the reported measures.

5.7 Supervised transfer versus zero-shot mapping

Both methods rank principles for the same 6,201 test examples. The MPNet definition-plus-anchor method reaches 67.63% Top-1 and 83.34% Top-2 agreement. The recovered supervised classifier reaches 93.00% and 97.68%. Table 7 gives the confidence intervals. The paired Top-1 difference is +25.37 percentage points (95% CI: +24.02 to +26.71). The comparison combines model representation and access to labelled training. The output measures principle-label agreement.

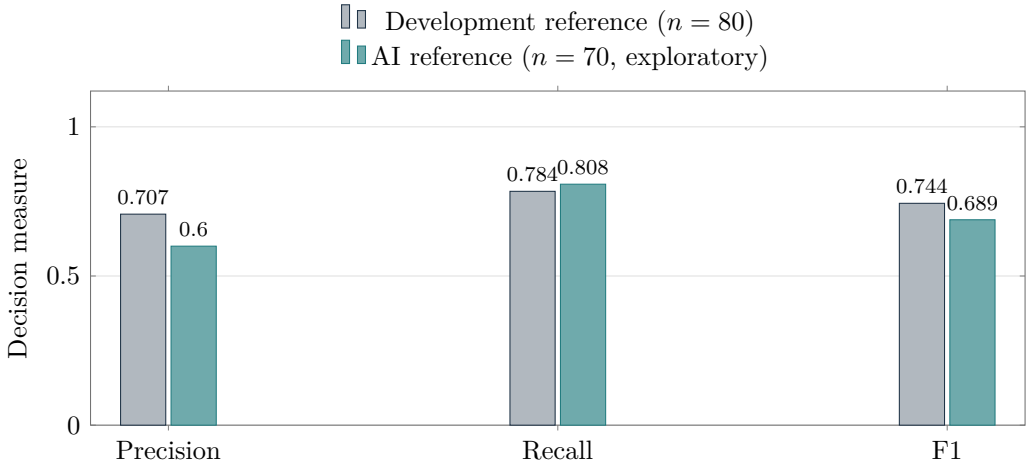


Figure 6. Precision, recall, and F1 for the fixed language detector. The first group contains 80 development examples; the second contains 70 AI-labelled examples. Precision describes how often a flag matches the reference, recall describes how many reference-vague passages are found, and F1 balances both. These measures cover the language branch.

Table 7. Principle mapping on the same company-separated test rows. Intervals resample companies. The supervised row reuses the classifier result and is not a second independent test.

Method	Top-1	Top-2	Macro-F1
MPNet zero-shot hybrid	67.63%	83.34%	0.6405
95% interval	66.45–68.93%	82.31–84.38%	0.6248–0.6567
Recovered supervised transfer	93.00%	97.68%	0.9158
95% interval	92.04–93.89%	97.10–98.21%	0.9036–0.9277

The supervised row reuses the classifier evaluation from the same company-separated test set. Both methods are compared against the filing-derived reference labels defined in Section 4.2. The result is reported once as an underlying model evaluation and reused for the mapping application.

5.8 Transparency Score compared with historical human ratings

The historical human-rating group contains 20 companies. Pearson correlation measures how automated and human scores vary together: $r = 0.6314$ (95% CI: 0.2683–0.8519; $p = 0.0028$). Spearman correlation compares score ranks: $\rho = 0.6285$ (95% CI: 0.2321–0.8540). Mean absolute error is 12.16 points, and root mean squared error, which gives larger errors more weight, is 15.19 points. The audit reports $r^2 = 0.3986$. This squared correlation describes association; no separate predictive regression was fitted for this result.

The historical comparison uses 20 company ratings from one documented source. The saved file contains both ratings and automated scores; rater blinding is not documented. The original Zensar text packet includes a URL-related extraction issue, and Geojit's business model differs from manufacturing-oriented reporting items. Both companies remain in the full group. The analysis uses the original ratings and fixed score formula, with all 20 cases retained; synthetic ratings and AI assessments are kept separate.

5.9 Checking the API connection

The audit records a successful principle-mapping API request for 360 ONE WAM LIMITED on 19 October 2026. The response identifies the recovered seed-99 classifier and checkpoint path. The

returned 89.17% summary is the average model probability for that filing. This request verifies model loading for one company.

6 Results Across Four Reporting Levels

The previous section evaluates the analysis methods. This section presents the saved outputs at four levels: one company, selected companies in one sector, the full financial-sector group, and all recorded sectors. All company values come from the saved exports. The examples connect passage-level analysis with company scores and sector summaries.

The saved database contains 1,164 company records and 1,172 filing records. The score export has one row for each of 1,164 unique CINs, using the most recent available filing selected by the script. These counts define the company collection used in the reporting examples.

6.1 One company: from filed facts to an interpretable outcome

Triveni Turbine Limited is a worked example with all four score components and a sector comparison. The linked database filing ends on 31 March 2026 and contains 1,862 raw facts. Table 8 shows three previous/current value pairs with matching concepts, units, and periods. Scope 1 emissions rise, Scope 2 emissions fall, and water withdrawal falls. These different directions show why one general claim of environmental improvement cannot be compared with every environmental figure.

The workflow stores the figures beside narrative facts and reporting categories. An eligible passage receives a principle label from an explicit tag, concept map, or principle note. The classifier ranks the nine principles, and the language detector identifies evidence markers and vague wording. Numerical checks retrieve the value pairs in Table 8. The worked example presents the saved facts and score components for this filing.

Every part of the saved score can be explained. The filing has 613 accepted concepts out of 1,057 in the database, giving Completeness of $100(613/1057) = 57.99\%$. Scoring retains 52 concept–text pairs. Specificity of 63.46% corresponds to 33 of 52 passages with a recognised evidence pattern. Clarity of 98.08% corresponds to one flagged passage and 51 without a flag. SDG breadth of 70.59% corresponds to 12 of the 17 goals under the fixed map. These counts reproduce the saved component percentages.

Equation (2) gives $23.196 + 19.038 + 19.616 + 7.059 = 68.909$, rounded to 68.91. Figure 7 plots each weighted contribution. The Machinery and Equipment group contains 36 companies with mean score 49.46. Triveni's percentile of 94.4 places the filing near the upper end of this recorded peer group.

Table 8. Filed values for Triveni Turbine, copied from filing record 197 without correction. Concept, period, and unit links allow the pairs to be checked. The source links identify the original company values.

Filed measure	FY2024–25	FY2025–26	Filed unit
Scope 1 emissions	4,142.84	5,177.03	tCO2e
Scope 2 emissions	3,353.61	3,169.15	tCO2e
Total water withdrawal	88,662.28	85,611.54	Kilolitres

Table 9. Transparency Score calculation for Triveni Turbine. Each percentage is multiplied by the weight. The final column shows the contribution before rounding the total.

Component	Percentage	Weight	Contribution
Completeness	57.99	0.40	23.196
Specificity	63.46	0.30	19.038
Clarity	98.08	0.20	19.616
SDG breadth	70.59	0.10	7.059
Transparency Score			**68.91**

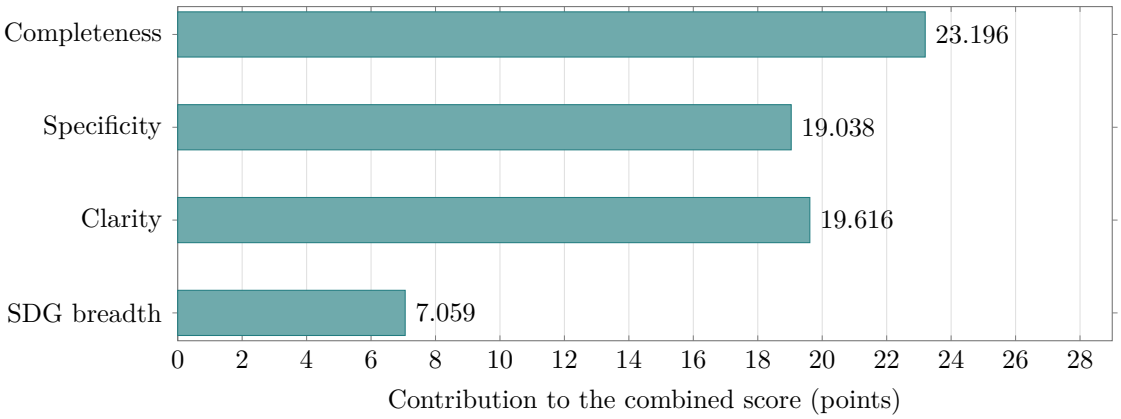


Figure 7. How the Triveni Turbine score is calculated. Each bar is the component percentage multiplied by the fixed weight. Completeness adds 23.196 points, Specificity 19.038, Clarity 19.616, and SDG breadth 7.059. The total is 68.909, rounded to 68.91. The bars show each component's contribution to the combined score.

6.2 Selected financial companies: component differences and available years

The second level compares companies within one recorded sector. The saved *Financial Services (excl. Insurance)* group has 94 companies. Table 10 shows the four highest and three lowest scores. These examples show the range and are not a random sample. The next level summarises the full group.

REC Limited scores 67.56 and Vardhman Holdings scores 29.83. Specificity is 57.69% for REC and 2.78% for Vardhman; Completeness is 56.01% and 35.00%. The component view shows where the disclosures differ under the same formula. Review also uses the business model, recorded sector, applicable reporting items, and available filing year to interpret the comparison.

Table 10. Financial-company examples from the high and low ends of the score range. FY end 2026 and 2025 mean 31 March of those years. C, Sp, Cl, and G are percentages; TS is the combined score. Years were reconstructed from the latest filing for each CIN.

Company	FY end	C	Sp	Cl	G	TS
REC	2026	56.01	57.69	98.08	82.35	67.56
Mahindra & Mahindra Financial Services	2026	67.83	49.15	98.31	29.41	64.48
YES BANK	2026	60.45	46.67	92.00	70.59	63.64
KOTAK MAHINDRA BANK	2026	61.21	59.42	95.65	17.65	63.21
Fedbank Financial Services	2025	31.98	12.12	75.76	0.00	31.58
Nalwa Sons Investments	2025	31.13	10.26	74.36	0.00	30.40
Vardhman Holdings	2025	35.00	2.78	75.00	0.00	29.83

Table 11. Two-year Transparency Scores for all eight available companies. Changes and weighted contributions are score points. C, Sp, Cl, and G are defined in Eq. (2). The last column gives the component with the largest absolute contribution to the change.

Company	FY2024–25	FY2025–26	Change	Largest contribution
Praj Industries	38.55	50.30	+11.75	C: +9.91
ESAF Small Finance Bank	42.59	54.10	+11.51	C: +8.10
Sudarshan Colorants	49.74	58.78	+9.04	Sp: +7.00
Gopal Snacks	41.30	47.29	+5.99	Sp: +8.53
G R Infraprojects	35.94	38.87	+2.93	Sp: +3.07
Sudarshan Chemical	62.95	65.32	+2.37	G: +2.35
VIP Industries	45.91	45.56	-0.35	Cl: -2.36
SML Mahindra	44.38	42.98	-1.40	Cl: -1.17
Group mean	45.17	50.40	+5.23	—

Table 12. All 94 financial companies grouped by score. Each company appears in one band. Component columns give mean percentages; TS gives the mean combined score. Supplementary Table S1 lists every company, reconstructed filing year, and sector percentile.

Score band	N	Mean TS	Range	C	Sp	Cl	G
60.00–69.99	11	62.39	60.01–67.56	61.17	47.24	95.25	47.06
50.00–59.99	35	54.77	50.23–59.84	60.69	32.94	94.07	17.98
40.00–49.99	26	45.22	40.57–49.96	45.85	26.05	89.79	11.08
Below 40.00	22	34.05	29.83–39.80	31.33	14.30	84.02	4.28
All financial companies	94	48.17	29.83–67.56	49.77	28.35	90.67	16.27

A year comparison uses repeated filings for the same company. For ESAF Small Finance Bank, Completeness rises from 34.34% to 54.59%, Specificity from 30.23% to 41.30%, and Clarity from 93.02% to 93.48%. SDG breadth stays at 11.76%. The combined score rises from 42.59 to 54.10, or 11.51 points. Completeness contributes about 8.10 points, Specificity 3.32 points, and Clarity 0.09 points. The breakdown connects each component change to the reported score movement.

Table 11 presents all eight companies with two available years. Six scores rise and two fall. The group mean rises from 45.17 to 50.40, a change of 5.23 points. The final column gives the component with the largest absolute weighted contribution to each change. Contributions from other components complete the total. Supplementary Table S3 retains both filing records, year labels, and every component for the eight-company group.

6.3 The complete financial-sector group

For all 94 financial-sector companies, the mean is 48.17, the median is 49.79, and the standard deviation is 9.88. Scores range from 29.83 to 67.56, a spread of 37.73 points. Mean Completeness is 49.77%, Specificity 28.35%, Clarity 90.67%, and SDG breadth 16.27%. Clarity is much higher than Specificity. This difference shows why avoiding vague phrases alone does not provide concrete evidence.

The latest recorded filing ends in March 2026 for 77 companies and March 2025 for 17. Year labels were reconstructed by CIN from the database using the script's latest-filing rule; the original score CSV omits a filing identifier. The group is a latest-available filing collection. The separate eight-company comparison uses paired reports to examine year-wise score changes.

Table 12 groups the scores into four ranges. Eleven companies score at least 60, 35 score from 50

to below 60, 26 score from 40 to below 50, and 22 score below 40. Mean Specificity is 14.30% in the lowest range and 47.24% in the highest. These ranges describe the distribution of the combined score and the associated component means.

The full financial group provides a score distribution and a component breakdown for all 94 companies. The financial mean is 0.56 points below the overall mean of 48.73. This summary places the sector alongside the full company collection, while the company rows show the variation within the sector.

6.4 All sectors: coverage, score distribution, and interpretation

The final level covers 1,164 company scores across 72 recorded sectors. Scores range from 26.88 to 75.58, with mean 48.73 and median 47.25. Forty-seven sectors contain at least five companies and account for 1,109 scores. The other 25 sectors contain 55 companies. Table 13 lists the 47 larger groups separately and combines the smaller groups in one labelled row. Supplementary Table S2 lists all individual sectors.

Among the 47 sectors used for comparison, Warehousing and Transport Support has the highest mean, 54.55 across six companies. Wearing Apparel has the lowest mean, 42.33 across six companies. Education follows at 42.43, also across six companies. The highest and lowest means differ by 12.22 points. These are descriptive comparisons of the saved company scores, with group sizes shown beside the means.

The larger groups provide further context. Chemicals Manufacturing has 101 companies with mean 49.56; Financial Services has 94 with mean 48.17; Pharmaceuticals has 65 with mean 51.18. Pharmaceuticals spans 32.38–73.72, and Chemicals spans 32.18–75.34. Reporting both sector means and company ranges presents the differences within each recorded group.

The Pharmaceuticals example gives another view within one sector. Dabur India scores 73.72 and Kopran scores 32.38, a gap of 41.34 points. Completeness is 73.13% and 32.92%, and Specificity is 57.38% and 23.68%. Clarity exceeds 85% for 58 of the 65 companies. The combined view shows how several components contribute to company rankings. Supplementary Table S4 preserves the four highest and three lowest examples with every component.

The four levels let an analyst move from a component calculation to a company score, then to a sector comparison, and finally to the full company collection. At each step, the denominator, source records, and meaning of the output remain visible. This structure supports both detailed disclosure review and summary comparison.

Table 13. Sector scores for all 1,164 companies. The 47 sectors with at least five companies are listed separately. The other 25 sectors are combined below the line. SD shows the spread of company scores. The main comparison uses a five-company reporting threshold. Supplementary Table S2 lists all 72 sectors and medians.

Sector or reporting group	N	Mean TS	Range	SD
Chemicals Manufacturing	101	49.56	32.18–75.34	9.32
Financial Services (excl. Insurance)	94	48.17	29.83–67.56	9.88
Pharmaceuticals	65	51.18	32.38–73.72	10.06
Basic Metals	60	50.80	34.61–75.58	10.87
Electrical Equipment	51	48.80	26.88–67.03	9.56
Computer Programming and Information Technology Services	49	50.24	32.89–73.98	12.25
Non-Metallic Mineral Products	40	50.36	35.66–69.20	8.69
Food Products Manufacturing	38	47.57	30.04–69.86	9.60
Motor Vehicles	37	51.76	33.29–70.12	9.98
Machinery and Equipment	36	49.46	35.61–69.67	10.06
Rubber and Plastics	36	49.22	35.44–73.87	9.90
Auxiliary Financial Services	31	47.26	27.67–61.39	9.08
Electricity, Gas, Steam and AC Supply	31	53.67	31.95–74.23	11.49
Fabricated Metal Products	27	50.29	35.03–65.72	8.66
Textiles	26	47.44	31.77–67.65	10.16
Wholesale Trade	24	45.46	28.05–64.86	12.39
Civil Engineering	23	48.31	31.99–66.15	10.84
Retail Trade	21	43.06	34.22–56.45	6.12
Construction	20	48.73	36.58–63.25	8.32
Insurance and Pension Funding	20	51.52	29.92–64.17	10.12
Real Estate Activities	19	47.90	35.17–63.42	8.78
Human Health Activities	18	46.37	33.92–62.86	8.12
Information Service Activities	18	47.39	33.97–64.19	9.71
Computer, Electronic and Optical Products	17	47.46	32.93–65.45	7.59
Other Transport Equipment	17	50.83	37.67–66.89	9.39
Other Manufacturing	16	46.11	30.59–63.81	10.19
Telecommunications	15	50.63	36.57–69.19	11.43
Accommodation	14	48.67	31.28–67.57	10.06
Mining and Quarrying	13	47.40	29.01–73.16	13.27
Beverages Manufacturing	10	47.30	34.70–67.71	13.18
Other (Division 99)	10	48.39	34.80–73.44	12.12
Wholesale/Retail Trade - Motor Vehicles	10	49.25	37.40–67.92	11.47
Architecture and Engineering	9	48.50	37.49–64.86	10.06
Coke and Refined Petroleum	9	52.32	38.20–69.63	11.13
Paper Products	9	43.27	33.42–48.93	4.88
Broadcasting	8	43.65	31.66–52.74	7.64
Land Transport	7	47.42	34.72–63.63	10.11
Water Supply and Waste Management	7	43.33	36.73–56.76	8.29
Wood Products	7	46.18	36.16–64.51	9.72
Education	6	42.43	33.28–55.70	7.74
Furniture	6	51.28	36.17–65.29	9.25
Other Professional/Technical Activities	6	48.43	38.73–57.90	8.70
Scientific R&D	6	44.90	38.95–52.11	5.71
Warehousing and Transport Support	6	54.55	37.82–67.03	10.34
Wearing Apparel	6	42.33	37.17–48.78	4.50
Leather Products	5	50.00	42.14–58.86	7.80
Water Transport	5	48.37	38.70–61.22	9.75
Other 25 small sectors (combined)	55	44.08	27.56–70.97	8.95
All 72 sectors	**1,164**	**48.73**	**26.88–75.58**	**9.98**

7 Discussion and Practical Use

7.1 How the outputs work together

GRASP connects three levels of analysis. Passage-level outputs identify likely principles and language flags. Filing-level outputs organise reported figures and calculate the four score components. Company and sector views then place those outputs within a broader reporting context. This structure lets an analyst move from a summary score to the underlying passage or filed item.

The classifier, language detector, and Transparency Score serve different purposes. The classifier ranks regulatory principles against filing-derived labels. The language detector selects passages for wording review. The score combines reported-item coverage, evidence markers, language flags, and mapped SDG breadth. The evaluation in Section 5 reports the data group and reference source for each output.

7.2 Reading the evaluation results

Source-wise results show how the classifier responds to explicitly tagged passages, concept-mapped passages, and principle notes. Tests on excluded concepts add a second view by changing which reporting questions appear in labelled training. Together, these comparisons describe model behaviour beyond the overall agreement value.

The company-separated split defines the labelled training and test groups. The checkpoint checks establish which saved parameters produced the reported results. Bootstrap intervals describe variation under company resampling, and paired comparisons evaluate both methods on the same selected companies. These recorded conditions make the comparisons clear and repeatable [11]. The principle-mapping result reuses the classifier evaluation, while the API check records successful model loading for one company.

7.3 Reading company and sector scores

The component breakdown explains how each company reaches the combined score. Completeness describes accepted responses across the fixed concept list. Specificity records evidence-marker presence. Clarity records passages without a vagueness flag, and SDG breadth counts goals linked through the configured principle map. Showing all four components lets an analyst identify the part of the disclosure that drives a score difference.

Sector comparisons use the saved industry groups and latest available filings. Year-wise comparisons use the eight companies with paired reports. These views describe the available company records under the same scoring rules. Detailed filing context, component values, and year labels support direct inspection of each comparison.

7.4 Use by analysts

An analyst can begin with a company or sector summary, select a component, inspect the source passage, and check the reported measure, period, and unit. Every flag retains the rule and source behind the decision. The service therefore supports a complete path from review priority to supporting material.

The four reporting levels also support different review tasks. A worked company example explains the calculation. Comparisons within a sector reveal different component patterns. The full financial group shows the score distribution, and the all-sector summary supports wider comparison. The numerical and assurance branches supply records for context review, while the reported evaluation identifies the specific component measured by each metric.

8 Conclusion

Large sustainability-report collections require a clear path from narrative claims to filed evidence and company-level summaries. GRASP provides this path through structured BRSR storage, filing-derived labels, principle classification, language checks, numerical review, and a four-part Transparency Score. Passages and figures remain linked to the same filing. Label sources remain visible during evaluation, and every score component has an explicit calculation. The resulting workflow combines passage review, company scoring, and sector comparison in one service.

The recovered classifier agrees with weak labels on 93.00% of test passages (95% CI: 92.04–93.89%) and reaches macro-F1 of 0.9158. The test set contains 6,201 passages from 175 companies excluded from supervised training. Agreement is 63.35% for explicitly tagged narratives, 95.15% for concept-mapped narratives, and 90.72% for principle notes. The fixed text-vagueness detector reaches F1 of 0.7436 on 80 development examples. The Transparency Score has Pearson correlation $r = 0.6314$ with 20 historical human ratings (95% CI: 0.2683–0.8519). These findings describe principle assignment, language screening, and score comparison under the evaluation conditions in Section 5.

The saved outputs cover 1,164 companies across 72 recorded sectors. Results move through four levels: one worked company, selected companies within a sector, the complete financial-sector group, and all sectors. Financial-company scores range from 29.83 to 67.56. Among the 47 sectors containing at least five companies, Wearing Apparel has the lowest mean, 42.33, and Warehousing and Transport Support has the highest, 54.55. Across the eight companies with paired-year scores, the mean rises by 5.23 points, from 45.17 to 50.40. Six scores rise and two fall. Component changes explain these movements within the available group. Company, sector, and year views make the saved results easy to inspect from both detailed and summary perspectives.

This work can be extended using multiple levels. At the data level, collection can include annexures, charts, and additional annual filings. At the modelling level, expanded principle-labelled datasets and blind human-rating studies can support new model comparisons. Adaptation using declared training-company text can be evaluated on newly collected company groups. Business descriptions can support more detailed sector assignment and indicator-applicability rules. At the numerical-review level, extensions can compare measure definitions, units, periods, and business boundaries before generating review records. At the longitudinal level, additional annual reports can support repeated comparisons of the same companies under a recorded score version. At the framework level, the filing-based approach can be studied for international sustainability standards through explicit reporting-item mappings and separate evaluation in each setting.

References

[1] Securities and Exchange Board of India, "Business responsibility and sustainability reporting by listed entities," Circular SEBI/HO/CFD/CMD-2/P/CIR/2021/562, 10 May 2021. https://www.sebi.gov.in/legal/circulars/may-2021/business-responsibility-and-sustainability-reporting-by-listed-entities_50096.html.

[2] N. Webersinke, M. Kraus, J. A. Bingler, and M. Leippold, "ClimateBERT: A pretrained language model for climate-related text," arXiv:2110.12010, 2021. https://arxiv.org/abs/2110.12010.

[3] J. Ni et al., "CHATREPORT: Democratizing sustainability disclosure analysis through LLM-based tools," in *Proceedings of EMNLP: System Demonstrations*, 2023, pp. 21–51. https://doi.org/10.18653/v1/2023.emnlp-demo.3.

[4] Y. Zou et al., "ESGReveal: An LLM-based approach for extracting structured data from ESG reports," *Journal of Cleaner Production*, vol. 489, article 144572, 2025. https://doi.org/10.1016/j.jclepro.2024.144572.

[5] T. Calamai, O. Balalau, T. Le Guenedal, and F. M. Suchanek, "Corporate greenwashing detection in text—a survey," arXiv:2502.07541, 2025. https://arxiv.org/abs/2502.07541.

[6] A. Garg, K. Upadhyay, D. Arora, V. Avashia, J. Shah, and V. Gupta, *Analysis and Insights from ESG Disclosures Submitted by 1012 Indian Businesses under BRSR Guidelines (2022–23)*. Indian Institute of Management Ahmedabad, NIIF Chair in ESG, 2025. https://www.iima.ac.in/sites/default/files/2025-02/BRSR%20Report_2-12-2024.pdf.

[7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186. https://doi.org/10.18653/v1/N19-1423.

[8] S. Gururangan et al., "Don't stop pretraining: Adapt language models to domains and tasks," in *Proceedings of ACL*, 2020, pp. 8342–8360. https://doi.org/10.18653/v1/2020.acl-main.740.

[9] Nils Reimers and Iryna Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proceedings of EMNLP-IJCNLP*, 2019, pp. 3982–3992. https://doi.org/10.18653/v1/D19-1410.

[10] A. Ratner, S. H. Bach, H. Ehrenberg, J. Fries, S. Wu, and C. Ré, "Snorkel: Rapid training data creation with weak supervision," *Proceedings of the VLDB Endowment*, vol. 11, no. 3, pp. 269–282, 2017. https://doi.org/10.14778/3157794.3157797.

[11] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, article 100804, 2023. https://doi.org/10.1016/j.patter.2023.100804.

[12] F. S. Varini, J. Boyd-Graber, M. Ciaramita, and M. Leippold, "ClimaText: A dataset for climate change topic detection," arXiv:2012.00483, 2020. https://arxiv.org/abs/2012.00483.

[13] J. A. Bingler et al., "Cheap talk and cherry-picking: What ClimateBERT has to say on corporate climate risk disclosures," *Finance Research Letters*, vol. 47, article 102776, 2022. https://doi.org/10.1016/j.frl.2022.102776.

[14] J. A. Bingler, M. Kraus, M. Leippold, and N. Webersinke, "How cheap talk in climate disclosures relates to climate initiatives, corporate emissions, and reputation risk," *Journal of Banking & Finance*, vol. 164, article 107191, 2024. https://doi.org/10.1016/j.jbankfin.2024.107191.

[15] T. Schimanski, J. Bingler, M. Kraus, C. Hyslop, and M. Leippold, "ClimateBERT-NetZero: Detecting and assessing net zero and reduction targets," in *Proceedings of EMNLP*, 2023, pp. 15745–15756. https://doi.org/10.18653/v1/2023.emnlp-main.975.

[16] A. Vinella, M. Capetz, R. Pattichis, C. Chance, R. Ghosh, and K.-W. Chang, "Leveraging language models to detect greenwashing," arXiv:2311.01469, revised 2024. https://arxiv.org/abs/2311.01469.

[17] R. Shankar and Q. Xu, "Automated detection of greenwashing in Indian corporate sustainability reports using natural language processing," SSRN working paper 5294480, posted 2025; manuscript dated 30 June 2024. https://doi.org/10.2139/ssrn.5294480.

[18] XBRL International, "Extensible Business Reporting Language (XBRL) 2.1," Recommendation, corrected errata, 20 February 2013. https://specifications.xbrl.org/work-product-index-group-base-spec-base-spec.html.

[19] K. Song, X. Tan, T. Qin, J. Lu, and T.-Y. Liu, "MPNet: Masked and permuted pre-training for language understanding," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 16857–16867. https://arxiv.org/abs/2004.09297.

[20] United Nations, "The 17 goals," Sustainable Development Goals, 2030 Agenda, 2015. https://sdgs.un.org/goals.